Wednesday Lab

1. A. Write a program in C to input a number and check whether the number is prime number or not using for loop. How to check prime numbers using loop in C programming. Logic to check prime numbers in C programming.
   Example Input: Input any number: 17
   Output: 17 is prime number

   B. Write a C program to print Fibonacci series up to n terms using loop. How to generate Fibonacci series up to n terms using loops in C programming. Logic to print Fibonacci series in a given range in C programming.
   Fibonacci Formula: F(n)=F(n−1)+F(n−2)
   Where: • F(n) is the nth Fibonacci number.
   • F(n−1) is the previous Fibonacci number.
   • F(n−2) is the Fibonacci number before the previous one.
   Initial Conditions:
   • F(0)=0
   • F(1)=1
   Example
   Input: Input number of terms: 10
   Output: Fibonacci series: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34

2. Problem Description: triangular number sequence is a sequence of numbers that can be arranged in the shape of an equilateral triangle. Write a C program to display the Triangular sequence up to n terms entered by the user.
   Tn=n×(n+1)/2

   Where:

   - Tn is the nth triangular number.
   - n is the term number.

   For example:

   - $T_1 = \frac{1\times(1+1)}{2} = 1$
   - $T_2 = \frac{2\times(2+1)}{2} = 3$
   - $T_3 = \frac{3\times(3+1)}{2} = 6$
   - $T_4 = \frac{4\times(4+1)}{2} = 10$

   Example:
   Input: Enter the number of terms 12
   Output: 0, 1, 3, 6, 10, 15, 21, 28, 36, 45, 55, 66

3. Problem Description: Write a C program that take n input numbers using scanf. And find the Maximum and minimum number among them and print the maximum and minimum value. (don;t use array)
   Example: Size of the list - 11
   Input the list - 8,4,12,34,56,89,7,34,11,33,54
   Output: The Maximum is 89. The minimum is 4.

4. Write a C program that prints a left-aligned pyramid of prime numbers. The pyramid should have N rows, where N is entered by the user. The first row should have 1 prime number, the second row 2 prime numbers, and so on. The prime numbers should be aligned to the left, creating a pyramid shape.

   Example:
   Input: Enter the number of rows: 5
   Output:
   2
   3 5
   7 11 13
   17 19 23 29
   31 37 41 43 47